

Computer Networks

Solved Problems on Delay

(Transmission, Propagation, Queuing and Processing delay)

1. Suppose that the packet length is 4000 bits, and that the link transmission rate along the link from the first router to the second is 10 MBps. What is the transmission delay?

$$L = 4000 \text{ bits}$$

$$R = 10 \text{ MBps} = 10 \times 8 \times 10^6 \text{ bps}$$

$$\begin{aligned} \text{Transmission Delay} &= L/R = 4000 / (10 \times 8 \times 10^6) \\ &= 0.00005 \text{ s} = 50 \text{ microseconds} \end{aligned}$$

2. The one hop transmission delay for a packet through a twisted pair cable between points A and B is 0.2s. What happens to the transmission delay (increases, decreases, no change) if:

- B is moved to a point C, 500 m further away from A.

No change

- The cable is changed to a coaxial cable

Decreases.

(In general, coaxial cables have 80x more transmission capacity than twisted-pair cables)

$$(d \propto 1/R)$$

- The bandwidth of the network is increased 5 fold.

Decreases.

$$(d \propto 1/R)$$

- The data being transferred is clumped into larger packets.

Increases.

$$(d \propto L)$$

3. A cable is able to transfer 5 packets each of 500 bits length in 0.4 seconds into the link connecting one router to another. What is the transmission delay?

$$\text{Transmission Delay} = L/R$$

$$L = 5000 \text{ bits}$$

$$R = \text{Bits transmitted per second}$$

$$= \text{Total bits transmitted} / \text{Total time taken}$$

$$= (5 \times 5000) / 0.4$$

$$\text{Transmission Delay} = 500 / ((5 \times 5000) / 0.4) = 0.4 / 5 = 0.08 \text{ s}$$

OR

$$\text{Transmission delay} = \text{time taken to put only one packet into the link}$$

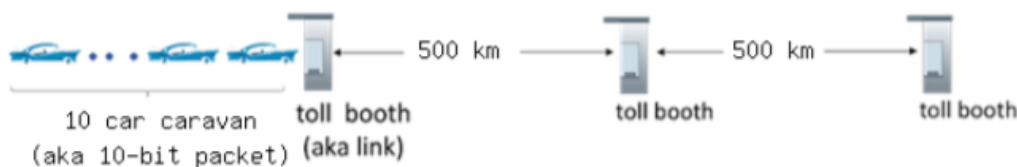
$$= 0.4 / 5 = 0.08s$$

4. There are two ways you can send a 10TB of data from Bengaluru to Chennai. You have a dedicated link for this very purpose transmitting at 100Mbps and a special courier option with delivery in one day. Which would you prefer to use?

10 TB = $10 * 10^{12} * 8$ bits.

Using the dedicated link, it will take $10 * 10^{12} * 8 / (100 * 10^6) = 80000$ seconds ~ 9 days. But with the special courier option, it would arrive in a single day. Hence the courier option would be preferred for speedy delivery.

5. Consider the figure below, which draws the analogy between store-and-forward link transmission and propagation of bits in packet along a link, and cars in a caravan being serviced at a toll booth and then driving along a road to the next toll booth.



Suppose the caravan has 10 cars, and that the toll booth services a car (that is, transmits) at a rate of one car per 5 seconds. Once receiving service, a car proceeds to the next toll booth, which is 500 kilometers away at a rate of 20 kilometers per second. Also assume that whenever the first car of the caravan arrives at a toll booth, it must wait at the entrance to the toll booth until all of the other cars in its caravan have arrived, and lined up behind it before being serviced at the toll booth. (That is, the entire caravan must be stored at the toll booth before the first car in the caravan can pay its toll and begin driving towards the next toll booth).

- a. Once the first car leaves the toll booth, how long does it take until it arrives at the next toll booth?

It takes 25 seconds to travel to the next toll booth (500 km / 20 km/s)

- b. Once the last car leaves the toll booth, how long does it take until it arrives at the next toll booth?

Just like in the previous question, it takes 25 seconds, regardless of the car

- c. Once the first car leaves the toll booth, how long does it take until it enters service at the next toll booth?

It takes 70 seconds until the first car gets serviced at the next toll booth ($10-1$ cars * 5 seconds per car + 500 km / 20 km/s)

- d. Are there ever two cars in service at the same time, one at the first toll booth and one at the second toll booth?

No, because cars can't get service at the next tollbooth until all cars have arrived

- e. Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first toll booth but not yet arrived at the second toll booth?

Yes, one notable example is when the last car in the caravan is serviced but is still traveling to the next toll booth; all other cars have to wait until it arrives, thus no cars are being serviced.

6. An optical fiber cable is able to transfer packets across the network at a speed of 70% of the speed of light in 10 microseconds. What is the distance the packet traverses?

$$\begin{aligned}\text{Speed of propagation} &= 70\% \text{ of } 3 \times 10^8 \text{ m/s} \\ &= 0.7 \times 3 \times 10^8 \\ &= 2.1 \times 10^8 \text{ m/s}\end{aligned}$$

$$\text{Propagation delay} = d/s$$

$$\begin{aligned}\text{Distance traversed} &= \text{Propagation delay} \times \text{speed of propagation} \\ &= 2.1 \times 10^8 \times 10 \times 10^{-6} = 2100 \text{ m}\end{aligned}$$

7. The time taken by a packet to travel across a link is 1000 times the time taken for the packet to be inserted in the link. Assuming no queuing and processing delay, what is the transmission delay of the link if it connects an end to end distance of 2km in the network in a total of 3 microseconds ?

$$\begin{aligned}\text{Speed of transmission } s &= 1000 \times \text{Transmission delay} \\ \text{Distance } d &= 2000 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Assuming no queuing and processing delay,} \\ \text{End to end delay} &= \text{Transmission delay} + \text{Propagation delay} \\ 3 &= d_{\text{trans}} + 2000 / (1000 \times d_{\text{trans}}) \\ d_{\text{trans}}^2 - d_{\text{trans}} \times 3 + 2 &= 0\end{aligned}$$

Solving the equation,

d_{trans} can be either 2 microseconds or 1 microsecond.

8. The propagation delay faced by Bob when he sends a packet to Alice is 0.05s. He thinks if he increases the bandwidth of his network, he'll be able to send the packet to her faster. Is he right?

No. Propagation delay is independent of bandwidth. It only depends on distance d of the communicating peers and the speed of transmission s in the employed medium.

9. Queuing Delay is the only delay that does not vary. True or False?

False. Queuing delay is the only delay that varies. The processing delay, transmission delay, and propagation delay are constant, if the route is fixed.

10. Bob can't reduce the distance between Alice and himself to help transmit packets faster. So he is looking into different cables to help him. His research helped him discover that twisted pair cables, coaxial cables and optical fiber cables allow packet propagation at speeds of 50%, 65% and 70% of the speed of light respectively. Which medium should he use to make Alice get his packets faster?

Optical fiber cables, because ($d \propto 1 / \text{speed of transmission}$)

11. Charlie tries to load both Alice's and Bob's websites on his system. Alice's server is 14m away and Bob's server is 8m away from Charlie's PC. If Charlie is connected to Alice's server through unguided media and Bob's through twisted pair cable (assume speed of transmission at 50% of the speed of light), whose website will load first on Charlie's system?

Comparing the propagation delays for both cases, d/s ,

Propagation delay for Alice's site = $14 / (3 \times 10^8)$

Propagation delay for Bob's site = $8 / (0.5 \times 3 \times 10^8) = 16 / (3 \times 10^8)$

Alice's site will load first as the propagation delay is lesser.

12. Suppose two hosts, A and B, are separated by 5000 kilometers and are connected by a direct link of $R = 4$ Mbps. Suppose the propagation speed over the link is 2×10^8 m/s. What is the maximum number of bits that the link can contain at any point of time during propagation?

$d = 5000 \times 1000$ m

$d_{\text{prop}} = d/s$

Maximum number of bits during propagation = $R \times d_{\text{prop}} = 4 \text{ Mbps} \times (5000 \times 1000 / (2 \times 10^8))$
= 0.1 Mb

There can be a maximum of 10^5 bits at any point of time during propagation in the link.

13. A new transmission is going to happen between Alice and Bob for a message of 4 bytes. The link through which they are connected passes through a router and also can only allow a maximum of 100 bits to be transferred at a time. In that case, what is the queuing delay Bob would face due to the router when Alice sends him a message?

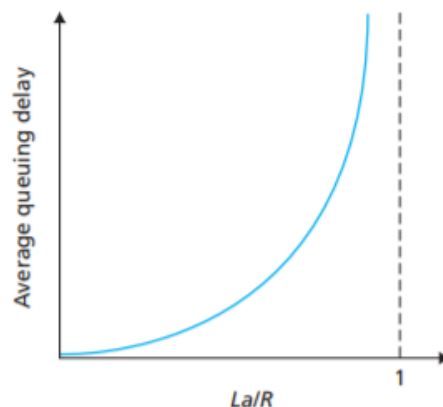
The entire message can fit in 1 packet and since no other packet has to be transmitted, there is no need for a queue. Therefore, the queuing delay would be 0.

14. Regarding queuing delay,
- If one packet arrives every L/R seconds, then what is the queuing delay?
 - If N packets arrive every $N(L/R)$ seconds, then what is the average queuing delay of the packets?

Then the first packet transmitted has no queuing delay; the second packet transmitted has a queuing delay of L/R seconds; and more generally, the N th packet transmitted has a queuing delay of $(N - 1)L/R$ seconds.

So, the average is
$$= (0 + L/R + 2L/R + \dots + (N-1)L/R) / N$$
$$= (N-1)L / 2R$$

15. Consider the queuing delay in a router buffer, where the packet experiences a delay as it waits to be transmitted onto the link. Traffic intensity $I = \lambda L/R$, and the queuing delay is calculated as $I(L/R)(1 - I)$ for $I < 1$ as shown in the below diagram.



- Assume a constant transmission rate of $R = 1700000$ bps, a constant packet-length $L = 1000$ bits, and λ is the average rate of packets/second which is 31. What is the queuing delay? Give your answer in milliseconds (ms)

$$\begin{aligned}\text{Traffic Intensity, } I &= \lambda a / R = 1000 \times 31 / 1700000 = 0.0182 \text{ packets} \\ \text{Queuing delay} &= I(L/R)(1 - I) \times 1000 \\ &= 0.0182 \times (1000/1700000) \times (1 - 0.0182) \times 1000 \\ &= 0.0105 \text{ ms.}\end{aligned}$$

- b. Assuming the router's buffer is infinite, the queuing delay is 0.0204 ms, and 522 packets arrive. How many packets will be in the buffer 1 second later?

$$\begin{aligned}\text{Packets leaving the buffer after 1 second} &= (1\text{s} / 0.0204 \times 10^{-3}\text{s}) \\ \text{Packets left in buffer} &= \text{arrived packets} - \text{packets gone} \\ &= 522 - \text{floor}(1 / 0.0204 \times 10^{-3}) = 0 \text{ packets. (cannot be negative)}\end{aligned}$$

- c. If the buffer has a maximum size of 500 packets, how many of the 522 packets would be dropped upon arrival from the previous question?

$$\text{Packets dropped} = \text{packets} - \text{buffer size} = 522 - 500 = 22 \text{ packets.}$$

16. Data traveling in Alice's network of 2 computers, takes a one hop transmission delay of 0.003ms to go from one computer to the other. If the queuing delay is 1 microsecond and the distance between the two systems is 3km, find the time taken by Alice's router for bit level error checking and processing if the total nodal delay is 0.04ms.

$$\begin{aligned}d_{\text{nodal}} &= d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}} \\ 0.04 \times 10^{-3} &= d_{\text{proc}} + 0.001 \times 10^{-3} + 0.003 \times 10^{-3} + 3000 / (3 \times 10^8) \\ \text{Processing delay, } d_{\text{proc}} &= (0.04 - 0.01 - 0.001 - 0.003) \times 10^{-3} = 0.026 \text{ ms}\end{aligned}$$

Problems on End-to-End Delay

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Question 1: 100 Tera Bytes of data is to be transferred between two hosts, Host A in Bengaluru and Host B in Chennai, separated by a distance of 350 km. The transmission speed at Host A is 1 Gbps and the propagation speed of the dedicated link is 2×10^8 m/s. Which one of these is better:

- to transfer the data over the link or
- to drive to Chennai to deliver the data and drive back to Bengaluru, if a one-way trip takes 7 hours

Which component is dominant in the total time taken for the data transmission? Assume negligible processing and queuing delays.

The end-to-end delay in transferring the data is given by

$$\begin{aligned}d_{end-end} &= d_{trans} + d_{prop} \\d_{end-end} &= \frac{L}{R} + \frac{d}{s} \\&= \frac{100 \times 10^{12} \times 8}{10 \times 10^9} + \frac{350 \times 10^3}{2 \times 10^8} \\&= 80000 + 0.00175 \\d_{end-end} &= 80000.00175 \text{ seconds} \approx 22.2 \text{ hours}\end{aligned}$$

It approximately takes 22 hours to transfer all the data between the hosts. On the other hand, it only takes 14 hours (2×7 hours) to deliver the data by hand.

Question 2: Host A is connected to Host B by two dedicated links, with transmission speeds 1 Gbps and 100 Mbps respectively. What is the minimum amount of time required for Host A to transfer 5.5 Giga Bytes of Data to Host B? Assume negligible queuing and propagation delays.

Host B can transfer the data efficiently by using both the links. Given the transmission speeds of the links, the amount of data transferred through each link must be divided in the ratio of their transmission speeds, which is 10 : 1. The total data to be transmitted through each link is

$$D_1 = \frac{10}{11} \times 5.5 = 5 \text{ GB}$$

$$D_2 = 5.5 - 5 = 0.5 \text{ GB}$$

The time taken to transmit the data through each link is

$$t_1 = \frac{5 \times 10^9 \times 8}{1 \times 10^9} = 40 \text{ seconds}$$

$$t_2 = \frac{0.5 \times 10^9 \times 8}{100 \times 10^6} = 40 \text{ seconds}$$

Hence, the minimum amount of time taken to transmit all of the data is 40 seconds (maximum of t_1 and t_2).

Question 3: Two hosts are separated by 10 nodes over a distance of 10,000 km. Starting from the first node, every other node has no queuing delay but has a processing delay of $0.5\mu s$ for each packet. Starting from the second node every other node has no processing delay but has a queuing delay of $1.5\mu s$ for each packet. There are no processing or queuing delays at the hosts. Assume the transmission speeds of the hosts and all the nodes is 100 Gbps. Also assume the propagation speed of all dedicated links is $2 \times 10^8 m/s$. What is the total time taken to transmit 10,000 packets each of size 64 Kilo Bytes between the hosts?

The total end-to-end delay is given by

$$\begin{aligned}
 d_{end-end} &= d_{proc} + d_{que} + d_{trans} + d_{prop} \\
 d_{proc} &= 5 \times (0.5 \times 10^{-6}) \times 10000 = 25 \text{ ms} \\
 d_{que} &= 5 \times (1.5 \times 10^{-6}) \times 10000 = 75 \text{ ms} \\
 d_{trans} &= \frac{L}{R} = \frac{10000 \times 64 \times 10^3 \times 8}{100 \times 10^9} = 51.2 \text{ ms} \\
 d_{prop} &= \frac{d}{s} = \frac{10000 \times 10^3}{2 \times 10^8} = 0.05 \text{ s} = 50 \text{ ms} \\
 d_{end-end} &= 25 + 75 + 51.2 + 50 = 201.2 \text{ ms}
 \end{aligned}$$

Question 4: Suppose a huge file of size 500 Giga Bytes is to be transferred over a distance of 100,000 km. The file is broken into packets of size 64 Bytes each. If the transmission speed at the host is 8.192 Gbps and the propagation speed of the dedicated link is $2 \times 10^8 m/s$, find the number of packets the link will contain at any point in time.

As the propagation speed of the link is $2 \times 10^8 m/s$, for a packet to reach the destination, it will take

$$d_{prop} = \frac{d}{s} = \frac{100000 \times 10^3}{2 \times 10^8} = 0.5 \text{ s} = 500 \text{ ms}$$

In 500 ms, the host can transmit

$$D = 8.192 \times 0.5 = 4.096 \text{ Gb}$$

4.096 Giga bits of data can be broken down into

$$\frac{4.096 \times 10^6}{64 \times 8} = 8 \times 10^6 \text{ packets}$$

Therefore, the link will contain 8×10^6 packets at any given time.

Question 5: Hosts A and B are connected by a satellite, at distances 6400 km and 8000 km respectively from it. The uplink and downlink transmission speeds are 1 Mbps and 8 Mbps respectively and the propagation speed in air is $2.5 \times 10^8 m/s$. If A and B send each other a packet that is 64 Kilo Bytes in size, calculate the extra end-to-end delay B experiences. Assume there are no processing and queuing delays.

As both A and B must transmit and receive one packet, the total transmission speed is the same for both of them. The only difference arises due to the fact that B is further away from the satellite than A is. Therefore, the difference in propagation delay is

$$\Delta d_{prop} = \frac{\Delta d}{s} = \frac{(8000 - 6400) \times 10^3}{2.5 \times 10^8} = \frac{1600 \times 10^3}{2.5 \times 10^8} = 6.4 \text{ ms}$$

Question 6: Suppose you are watching a cricket match live on your computer. The provider offers a 1080p stream (1920×1080 pixels) at 24 frames per second. Each pixel requires 3 Bytes to represent the RGB color levels. If the processing, queuing and the propagation delays are absent and the data for the audio is negligible, what is the minimum transmission speed required at the server to have less than 1 second end-to-end delay for the stream?

The total amount of data being transmitted in one second is

$$D = 1920 \times 1080 \times 3 \times 8 \times 24 \approx 1.2 \text{ Giga bits}$$

We know

$$d_{trans} = \frac{L}{R}$$

To find R

$$R = \frac{L}{d_{trans}} = \frac{1.2 \times 10^9}{1} = 1.2 \text{ Gbps}$$

Therefore, the server must have a minimum transmission speed of 1.2 Gbps for an end-to-end delay of less than 1 second.

Question 7: An ISP offers internet connections with transmission speeds of 10 Mbps. To attract customers, it promises less than 5 ms of end-to-end delay while playing online games, if the other players are also their customers and are no more than 10,000 km away. Assume an idle network with propagation speed of $2.5 \times 10^8 \text{ m/s}$ and no processing and queuing delays. If the average amount of data generated per second is 100 Kilo Bytes, find out if the ISP can deliver on their promise.

The end-to-end delay in transferring the data is given by

$$\begin{aligned} d_{end-end} &= d_{trans} + d_{prop} \\ d_{end-end} &= \frac{L}{R} + \frac{d}{s} \\ &= \frac{100 \times 10^3 \times 8}{10 \times 10^6} + \frac{10000 \times 10^3}{2.5 \times 10^8} \\ &= 0.08 + 0.04 \\ d_{end-end} &= 0.12 \text{ seconds} = 120 \text{ ms} \end{aligned}$$

In the worst-case scenario, the end-to-end delay is 120 ms. Therefore, the ISP is lying.

Question 8: Suppose there are two networks, Network A and Network B, in mesh and star topology respectively. In Network A, the transmission speed of the hosts is 10 Mbps and the queuing delay of each packet is 5 ms. In Network B, the transmission speeds of the hosts and the hub are 100 Mbps and 1 Gbps respectively and the queuing delays are 5 ms and 1 ms respectively. If a packet of size 64 Kilo Bytes is to be sent from one host to another, which network has the least end-to-end delay? Assume negligible processing and propagation delays.

In network A, the end-to-end delay in transferring the data is given by

$$\begin{aligned} d_{end-end(A)} &= d_{que(host)} + d_{trans(host)} \\ d_{end-end(A)} &= (5 \times 10^{-3}) + \frac{64 \times 10^3 \times 8}{10 \times 10^6} \\ &= 5 + 51.2 \\ d_{end-end(A)} &= 56.2 \text{ ms} \end{aligned}$$

In network B, the end-to-end delay in transferring the data is given by

$$\begin{aligned} d_{end-end(B)} &= d_{que(host)} + d_{trans(host)} + d_{que(hub)} + d_{trans(hub)} \\ d_{end-end(B)} &= (5 \times 10^{-3}) + \frac{64 \times 10^3 \times 8}{100 \times 10^6} + (1 \times 10^{-3}) + \frac{64 \times 10^3 \times 8}{1 \times 10^9} \\ &= 5 + 5.12 + 1 + 0.512 \\ d_{end-end(B)} &= 11.632 \text{ ms} \end{aligned}$$

Hence, packet transmission is faster in Network B (star topology).

Question 9: The Iridium constellation is a constellation of low-earth-orbit satellites, with 12 satellites orbiting along the circumference of the earth at any given time. If a call, with one packet of 64 Kilo bits in size generated every second, is to be made from the South Pole to the North Pole, what is the end-to-end delay experienced? Assume no delays at the hosts, queuing delay of 5 ms per packet and transmission speed of 10 Mbps at the satellites and propagation speed of 3×10^8 m/s. Also assume the radius of the earth is 6400 km.

As the North Pole and South Pole are at opposite ends of the earth, the distance between them is half the circumference of the earth. Therefore, there are 6 satellites between them (on average) and the distance is

$$d = \pi \times 6400 \approx 20106 \text{ km}$$

The end-to-end delay in transferring the packet is given by

$$\begin{aligned} d_{end-end} &= d_{que} + d_{trans} + d_{prop} \\ d_{que} &= 6 \times 5 = 30 \text{ ms} \\ d_{trans} &= \frac{L}{R} = \frac{64 \times 10^3 \times 8}{10 \times 10^6} = 51.2 \text{ ms} \\ d_{prop} &= \frac{d}{s} = \frac{20106 \times 10^3}{3 \times 10^8} \approx 67 \text{ ms} \\ d_{end-end} &= 30 + 51.2 + 67 = 148.2 \text{ ms} \end{aligned}$$

The end-to-end delay experienced is 148.2 ms.

Question 10: Two hosts are separated by a distance of 5,000 km. Two packets that are 4 Mega Bytes in size must be sent from one host to the other. The queuing delay at the host is 10 ms and the the propagation speed of the link is 2.5×10^8 m/s. Find the transmission speed at the source such that it finishes transmitting the second packet when the first packet just arrives at the destination.

The propagation speed is given by

$$d_{prop} = \frac{d}{s} = \frac{5000 \times 10^3}{2.5 \times 10^8} = 20 \text{ ms}$$

Given,

$$d_{que} + d_{trans} = d_{prop}$$

To find d_{trans}

$$d_{trans} = d_{prop} - d_{que} = 20 - 10 = 10 \text{ ms}$$

We know

$$d_{trans} = \frac{L}{R}$$

To find R

$$R = \frac{L}{d_{trans}} = \frac{4 \times 10^4 \times 8}{20 \times 10^{-3}} = 3.2 \text{ Gbps}$$

Question 11: A network has 4 hosts arranged in the ring topology. The queuing delay at a host is 10 ms per packet and the transmission speed is 1 Gbps. If a packet of size 4 Mega Bytes is to be transmitted between any two hosts, what is the average end-to-end delay? Assume there are no processing and propagation delays.

Let the hosts in the network be A, B, C and D, such that B and C are adjacent to A and D is opposite A. The average end-to-end delay is given by

$$d_{end-end(avg)} = \frac{d_{end-end(A-B)} + d_{end-end(A-C)} + d_{end-end(A-D)}}{3}$$

$$\begin{aligned}
d_{end-end(A-B)} &= d_{end-end(A-C)} = d_{que} + d_{trans} \\
&= (10 \times 10^{-3}) + \frac{4 \times 10^6 \times 8}{1 \times 10^9} \\
d_{end-end(A-B)} &= 10 + 32 = 42 \text{ ms} \\
d_{end-end(A-D)} &= d_{que(A-B)} + d_{trans(A-B)} + d_{que(B-D)} + d_{trans(B-D)} \\
&= (10 \times 10^{-3}) + \frac{4 \times 10^6 \times 8}{1 \times 10^9} + (10 \times 10^{-3}) + \frac{4 \times 10^6 \times 8}{1 \times 10^9} \\
d_{end-end(A-D)} &= 10 + 32 + 10 + 32 = 84 \text{ ms}
\end{aligned}$$

Therefore,

$$d_{end-end(avg)} = \frac{42 + 42 + 84}{3} = 56 \text{ ms}$$

Question 12: 100 packets each of size 16 Kilo Bytes must be transmitted from a server to two hosts, A and B, each at a distance of 5,000 km from the server. Transmission to host A occurs first and after its completion, transmission to host B starts. The link connecting the server and host A has a packet loss probability of 20%, while that connecting host B has a packet loss probability of 10%, and each lost packet is re-transmitted. If each link has propagation speed $2 \times 10^8 \text{ m/s}$ and the transmission speed at the server is 1 Gbps, at what times are the transmissions to each host completed? Assume negligible processing and queuing delays.

Since the link connecting host A and the server has a packet loss probability of 20%, the total number of packets transmitted to host A is

$$100 + (20\% \text{ of } 100) = 120$$

The end-to-end delay in transferring the packets to host A is given by

$$\begin{aligned}
d_{end-end} &= d_{trans} + d_{prop} \\
d_{end-end} &= \frac{L}{R} + \frac{d}{s} \\
&= \frac{120 \times 16 \times 10^3 \times 8}{10 \times 10^9} + \frac{5000 \times 10^3}{2 \times 10^8} \\
d_{end-end} &= 15.36 + 25 = 40.36 \text{ ms}
\end{aligned}$$

As the link connecting host B and the server has a packet loss probability of 10%, the total number of packets transmitted to host A is

$$100 + (10\% \text{ of } 100) = 110$$

The end-to-end delay in transferring the packets to host B is given by

$$\begin{aligned}
d_{end-end} &= d_{trans} + d_{prop} \\
d_{end-end} &= \frac{L}{R} + \frac{d}{s} \\
&= \frac{110 \times 16 \times 10^3 \times 8}{10 \times 10^9} + \frac{5000 \times 10^3}{2 \times 10^8} \\
d_{end-end} &= 14.08 + 25 = 39.08 \text{ ms}
\end{aligned}$$

Transmissions to hosts A and B are completed after 40.36 ms and 79.44 ms ($40.36 + 39.08$) respectively.

Question 13: In the new 5G technology, the peak upload and download speeds are in the ratio 1:2. Find their values if a packet of size 2.5 Mega Bytes experiences an end-to-end delay of 3 ms. Assume the processing, queuing and propagation speeds are negligible.

$$\begin{aligned}
d_{end-end} &= d_{trans(upload)} + d_{trans(download)} \\
3 \times 10^{-3} &= \frac{2.5 \times 10^6 \times 8}{x} + \frac{2.5 \times 10^6 \times 8}{2x}
\end{aligned}$$

To find x

$$x = \frac{2.5 \times 10^6 \times 8 + 2 \times 2.5 \times 10^6 \times 8}{3 \times 10^{-3}}$$

$$x = \frac{2.5 \times 10^6 \times 8}{3 \times 10^{-3}} = 10 \times 10^9 = 10 \text{ Gbps}$$

Therefore, the peak upload and download speeds are 10 Gbps and 20 Gbps respectively.

Question 14: There are two paths to transmit 100 packets each of size 16 Mega Bytes from host A to host B: a lossy channel with loss probability of 50% and transmission speed of 20 Gbps; and a lossless channel with transmission speed of 10 Gbps. If both the channels have a propagation speed of $2 \times 10^8 \text{ m/s}$ and the queuing delay at the host is 10 ms per packet, which path is better? Assume all lost packets are re-transmitted with no additional overhead.

Since both paths have the same propagation speed and queuing delay, the only difference arises from the transmission delays. For the lossy channel, the total number of packets that must be transmitted is

$$100 + (50\% \text{ of } 100) = 150$$

The transmission delay in the lossy channel is

$$d_{trans} = \frac{L}{R} = \frac{150 \times 16 \times 10^6 \times 8}{20 \times 10^9}$$

$$d_{trans} = 960 \text{ ms}$$

For the lossless channel, the total number of packets that must be transmitted is 100. The transmission delay in the lossless channel is

$$d_{trans} = \frac{L}{R} = \frac{100 \times 16 \times 10^6 \times 8}{10 \times 10^9}$$

$$d_{trans} = 1280 \text{ ms}$$

Therefore, transfer of the packets is faster in the lossy channel.

Question 15: Suppose a circuit-switched network has a one-time setup delay of 500 ms and a transmission speed of 100 Mbps, with no additional queuing delay. Also, suppose a packet-switched network has a queuing delay of 10 ms per packet and transmission speed of 10 Mbps. After transmission of how many packets, each of size 16 Kilo Bytes, does the circuit-switched network seem like a better choice for data transmission in terms of end-to-end delay? Assume transmission is happening between hosts separated by the same distance and the propagation speed of the links is also the same in both the networks.

As it is given that the hosts in both the cases are separated by the same distance, the propagation time will be same. Hence, it need not be calculated. Let x packets be transmitted. In the circuit-switched network, the end-to-end delay is given by

$$d_{end-end(1)} = d_{setup} + d_{trans}$$

$$d_{end-end(1)} = 0.5 + \frac{L}{R}$$

$$= 0.5 + \frac{x \times 16 \times 10^3 \times 8}{100 \times 10^6} \text{ seconds}$$

In the packet-switched network, the end-to-end delay is given by

$$d_{end-end(2)} = d_{que} + d_{trans}$$

$$d_{end-end(2)} = 0.01x + \frac{L}{R}$$

$$= 0.01x + \frac{x \times 16 \times 10^3 \times 8}{10 \times 10^6} \text{ seconds}$$

For the circuit switched network to be more efficient,

$$d_{end-end(1)} < d_{end-end(2)}$$

$$\begin{aligned}
0.5 + \frac{x \times 16 \times 10^3 \times 8}{100 \times 10^6} &< 0.01x + \frac{x \times 16 \times 10^3 \times 8}{10 \times 10^6} \\
0.5 &< 0.01x + \frac{128x}{10^4} - \frac{12.8x}{10^4} \\
100x + 128x - 12.8x &> 5000 \\
x &> \frac{5000}{215.2} \\
x &\geq 24
\end{aligned}$$

When at least 24 packets are transmitted, the circuit-switched network becomes more efficient than the packet-switched network.

Computer Networks - Throughput Problems

1. Hosts Alice and Bob are connected to each other via a router. The bandwidth of each link between Alice and the router is 15Mbps. If the efficiency of data transfer in the entire network is 80%, what is the actual throughput of the network?

$$IdealThroughput = \min\{15, 15\} = 15Mbps$$

$$Efficiency = \frac{\text{Amount of data transferred through the network per second}}{\text{Amount of data that could be transferred per second}}$$

$$\frac{80}{100} = \frac{x}{15}$$

$$Throughput(x) = 12Mbps$$

2. A network with a bandwidth of 10 Mbps can pass only an average of 30,000 frames per minute where each frame carries an average of 10,000 bits. What will be the throughput for this network?

$$Throughput = \frac{30,000 * 10,000}{60}$$

$$Throughput = 5Mbps$$

The throughput is nearly equal to half of the bandwidth.

3. Consider a 10 Mbps link being shared by 7 users. What is the maximum throughput if the network is using using circuit switching?

$$Throughput = \frac{\text{Total link capacity}}{\text{No of users}}$$

$$Throughput = \frac{10}{7}$$

$$Throughput = 1.4Mbps$$

The throughput is nearly equal to half of the bandwidth.

4. Consider a 10 Mbps link college network link, and this link is being shared by 5 users. Suppose packet switching is used. What is the network throughput if the users are using 20% of the network bandwidth?

$$Throughput = 10Mbps$$

The network throughput is the maximum because 5 users using for 20% of the time is 100% usage.

5. Alice and Bob are connected to each other via a router. The bandwidth of each link between the router and Bob is 20Mbps. If the network allows for the actual transmission of data at 20Mbps, then what is the time taken for a packet of size 1000 bits to enter the network?

Let R = bandwidth for the link between Alice and the router

$$Throughput = \min\{20, x\}$$

$$Throughput = 20Mbps$$

$$Timetaken = \text{Transmission delay} = \frac{\text{Packet Length}}{R}$$

$$Timetaken = \frac{1000}{20 * 10^6} = 50microseconds$$

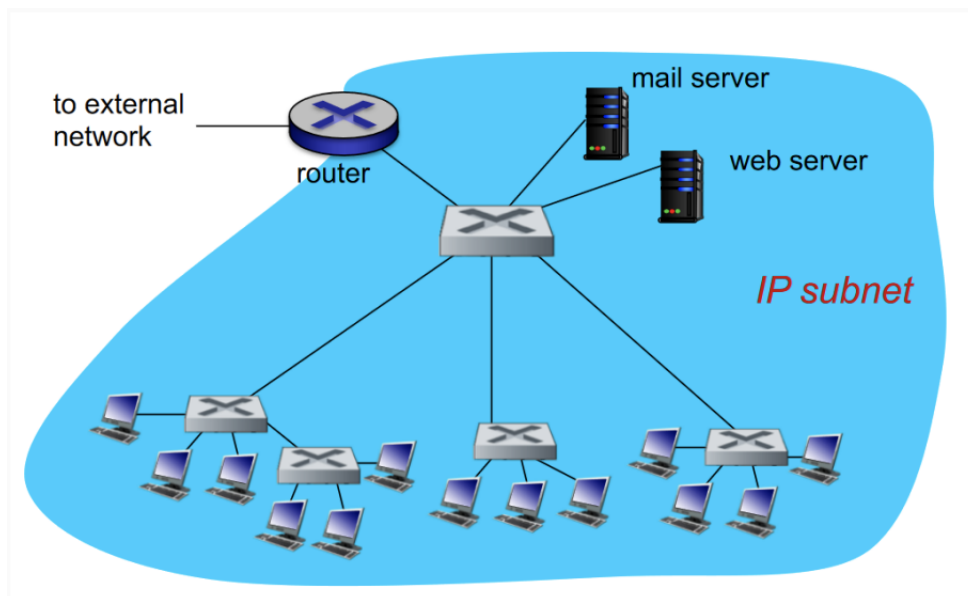
6. Your laptop is connected to the University WiFi network, you are downloading a 100kb file from the university file server. The WiFi link speed is 10Mbps and the server is connected to the network with a 100Mbps ethernet cable. What is the throughput while downloading the file onto your computer?

$$Throughput = \min\{R_{wifi}, R_{router}\}$$

$$Throughput = \min\{10, 100\}$$

$$Throughput = 10Mbps$$

7. What is the network throughput among the hosts and users of the below network, assuming that all the links are 150 Mbps?



If all 15 nodes are using the entirety of their link's bandwidth

$$Throughput = 15 * 100 = 1500Mbps$$

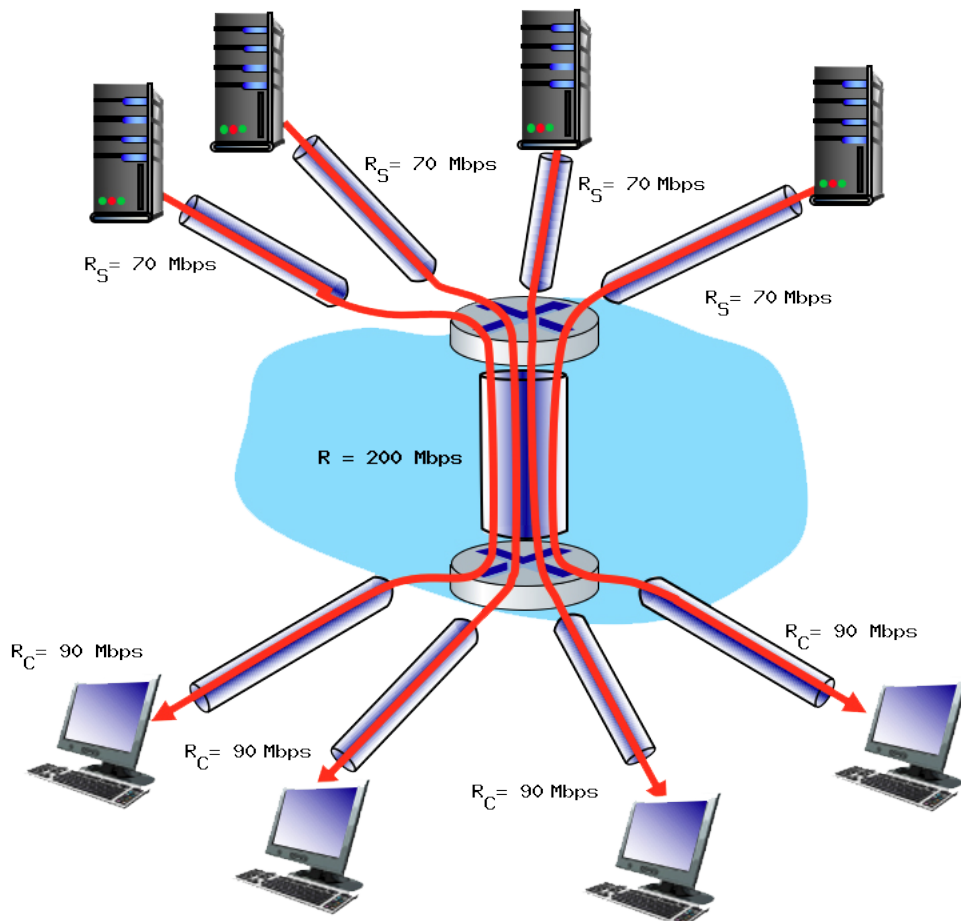
8. Consider the figure below where transmission delay is the only significant delay. Each link is 2Mbps. Suppose the number of links N is 3. Calculate the end to end delay if message of size 8 Mb is transferred and calculate the throughput for this scenario.

$$\begin{aligned} \text{Delay} \\ L &= 8Mb \\ \text{End to end delay} &= 3 * \frac{L}{R} = 3 * \frac{8}{2} = 12sec \\ \text{Throughput} &= \frac{8Mb}{12sec} = 0.667Mbps \end{aligned}$$

9. Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R_1 = 10$ Mbps, $R_2 = 5$ Mbps, and $R_3 = 30$ Mbps. Assuming no other traffic in the network, what is the throughput for the file transfer?

$$\begin{aligned} \text{Throughput} &= \min\{R_1, R_2, R_3\} \\ \text{Throughput} &= \min\{10, 5, 30\} \\ \text{Throughput} &= 5Mbps \end{aligned}$$

10. Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 200$ Mbps. The four links from the servers to the shared link have a transmission capacity of $R_S = 70$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 90$ Mbps.



- (a) What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

$$\text{Bottleneck} = \min\{R_c, R_s, R\}$$

$$\text{Bottleneck} = R = 200 \text{ Mbps}$$

Since there are 4 devices connected,

$$\text{Maximum throughput} = R/4 = 200/4 = 50 \text{ Mbps}$$

- (b) Which link is the bottleneck link?

R, since 50Mbps is the highest the middle hop can offer.

- (c) Assuming that the servers are sending at the maximum rate possible, what are the link utilization's for the server links (RS)? (In Percentage)

Assuming maximum rate at server as 70 Mbps, and bottleneck as 50 Mbps

$$\text{Utilization} = (50/70) * 100 = 71$$

- (d) Assuming that the servers are sending at the maximum rate possible, what are the link utilization's for the client links (RC)?

Assuming maximum rate at client as 90 Mbps, and bottleneck as 50 Mbps

$$\text{Utilization} = (50/90) * 100 = 56$$

- (e) Assuming that the servers are sending at the maximum rate possible, what is the link utilization's for the shared link (R)?

Assuming maximum rate at server as 70 Mbps, and bottleneck as 50 Mbps

Since there are 4 streams at the bottleneck, $50 * 4 = 200 \text{ Mbps}$

$$\text{Utilization} = (200/200) * 100 = 100$$